

plugin is used, which enables the client to send the password to the server as cleartext. There is no corresponding server-side plugin. Rather, `mysql_clear_password` can be used on the client side in concert with any server-side plugin that needs a cleartext password. (Examples are the PAM and simple LDAP authentication plugins; see [Section 6.4.1.5, “PAM Pluggable Authentication”](#), and [Section 6.4.1.7, “LDAP Pluggable Authentication”](#).)

The following discussion provides usage information specific to cleartext pluggable authentication. For general information about pluggable authentication in MySQL, see [Section 6.2.17, “Pluggable Authentication”](#).

**Note**

Sending passwords as cleartext may be a security problem in some configurations. To avoid problems if there is any possibility that the password would be intercepted, clients should connect to MySQL Server using a method that protects the password. Possibilities include SSL (see [Section 6.3, “Using Encrypted Connections”](#)), IPsec, or a private network.

To make inadvertent use of the `mysql_clear_password` plugin less likely, MySQL clients must explicitly enable it. This can be done in several ways:

- Set the `LIBMYSQL_ENABLE_CLEARTEXT_PLUGIN` environment variable to a value that begins with `1`, `Y`, or `y`. This enables the plugin for all client connections.
- The `mysql`, `mysqladmin`, `mysqlcheck`, `mysqldump`, `mysqlshow`, and `mysqlslap` client programs support an `--enable-cleartext-plugin` option that enables the plugin on a per-invocation basis.
- The `mysql_options()` C API function supports a `MYSQL_ENABLE_CLEARTEXT_PLUGIN` option that enables the plugin on a per-connection basis. Also, any program that uses `libmysqlclient` and reads option files can enable the plugin by including an `enable-cleartext-plugin` option in an option group read by the client library.

6.4.1.5 PAM Pluggable Authentication

**Note**

PAM pluggable authentication is an extension included in MySQL Enterprise Edition, a commercial product. To learn more about commercial products, see <https://www.mysql.com/products/>.

MySQL Enterprise Edition supports an authentication method that enables MySQL Server to use PAM (Pluggable Authentication Modules) to authenticate MySQL users. PAM enables a system to use a standard interface to access various kinds of authentication methods, such as traditional Unix passwords or an LDAP directory.

PAM pluggable authentication provides these capabilities:

- External authentication: PAM authentication enables MySQL Server to accept connections from users defined outside the MySQL grant tables and that authenticate using methods supported by PAM.
- Proxy user support: PAM authentication can return to MySQL a user name different from the external user name passed by the client program, based on the PAM groups the external user is a member of and the authentication string provided. This means that the plugin can return the MySQL user that defines the privileges the external PAM-authenticated user should have. For example, an operating system user named `joe` can connect and have the privileges of a MySQL user named `developer`.

PAM pluggable authentication has been tested on Linux and macOS.